When progressing training loads, what are the considerations for healthy and injured athletes?

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Every sport requires specific physical capacities to enable success and minimise injury risk. For instance, a competitive runner requires adequate muscular force production to counter ground reaction forces for a single step at a given speed (*local tissue* capacity), and the muscular and cardiovascular endurance to tolerate the accumulation of steps for the time and distance of the event (*sport-specific* capacity). Failing to adequately develop these physical qualities may result in underperformance and/or increased injury risk. In this editorial we explore how to optimally monitor and progress training loads to improve *local tissue* and *sport-specific* capacity.

UNDERSTANDING THE INTERPLAY BETWEEN *SPORT-SPECIFIC* AND *LOCAL TISSUE* CAPACITY

When referring to load capacity, *sport-specific* capacity is defined as the athlete's ability to perform (and withstand) the demands of training and competition, whereas *local tissue* capacity is defined as a specific structure's ability to withstand tissue-specific cumulative load.[1] In healthy athletes, training load quantification focuses on the athlete's *sport-specific* capacity and not on one's specific *local tissue* capacity. Progressively applied *sport-specific* training load improves an athlete's physical capacities (eg, strength, power and endurance) and performance.[2] Conversely, injury risk increases if training loads grossly exceed the athlete's current *local tissue* capacity.[3] Thus, increasing *local tissue* capacity as a component of the training cycle can decrease injury risk.

OPTIMAL LOADING OF *LOCAL TISSUE* CAPACITY AND *SPORT-SPECIFIC* CAPACITY FOR HEALTHY AND INJURED ATHLETES

Loading for healthy athletes is designed to improve *sport-specific* capacity, which in turn increases performance and may reduce injury risk. Although *sport-specific* training may enhance tissue-specific capacity (eg, sprint training for hamstrings), focused muscular loading is performed regularly to maintain *local tissue* capacity (eg, hamstring strength training).

For the injured athlete, *local tissue* capacity immediately decreases after acute injury, likely driven by tissue damage and pain, with psychosocial factors as effect mediators. In the early stages of injury, management may involve the temporary reduction of *local tissue* loads to promote healing and reduce irritability, followed by a restoration of *local tissue* capacity through progressive strengthening and improvements in neuromuscular control. Concomitant loss of tissue capacity can also occur in sites other than the original injury, as seen with bone strength in the contralateral limb 24 weeks post-tibial bone stress injury.[4] In both acute and persistent injury, neuromuscular deficits, a decline in tissue strength, and subsequent loss of *sport-specific* capacity may result.[4]

PROGRESSIVE LOADING IN THE TREATMENT OF MUSCULOSKELETAL INJURIES

Currently, treatment for musculoskeletal injuries involves progressive *local tissue* loading based on the patient's pain response, tissue vulnerability, movement pattern, re-injury risk, psychosocial resilience, temporal stage of tissue healing and inflammatory response to rehabilitation.[5] Excluding bone stress injuries, using pain solely as a progression guide provides poor feedback about the athlete's *local tissue* tolerance to the prescribed loads, as there is often a mismatch between pain, function and structure.[6] Therefore, loads that are within pain tolerance may not be sufficient for improvements in *local tissue-specific* load capacity and function. Importantly, restoring *local tissue* capacity in injured athletes is not enough to return athletes to play and perform safely. *Sport-specific* loading (eg, sprinting) results in greater muscle activation[7] and training-induced architectural changes[8] than targeted *local tissue* loading (eg, hamstring eccentric strength training). Clearly, maintenance or restoration of *sport-specific* capacity is required to bridge the gap between rehabilitation and peak performance (figure 1, online supplemental table 1).

HOW SHOULD *LOCAL TISSUE* LOADING BE QUANTIFIED?

In order to prevent underloading on reconditioning and return to sport, clinicians require a means of quantifying and progressing *local tissue* loading in injured athletes. The session-rating of perceived exertion (RPE) is the most commonly used approach to quantify the global internal training loads of athletes. Although differential RPE for breathlessness and leg-muscle exertion has been proposed as a method to enhance precision of internal load measures,[9] the difficulty in quantifying *local tissue* loads is a major limitation in current load monitoring practices. Is an RPE scale (commonly used for *sport-specific* loads) sensitive enough to capture internal load of the tissue? Does the visual analogue scale combined with external loads adequately capture the load-response required for optimal rehabilitation? Our ability to accurately measure tissue-specific loads performed during rehabilitation is still lacking.[10]

THREE KEY POINTS WHEN PROGRESSING TRAINING LOADS

1. Progressive loading using patient-reported feedback is best practice.
   Accurately quantifying *local tissue* loads in the injured athlete is difficult. The inclusion of a patient-reported outcome (such as a session-RPE or a visual analogue pain scale), in combination with external loads are needed to address *local tissue* loads and progress exercises safely and efficiently.
2. Effective programmes employ *local tissue* loading to maintain *local tissue* capacity.
   Local tissue loading should be prescribed with the healthy athlete's previous injury(ies) in mind, since prior

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In healthy athletes, the prescription of training load prioritises the development of sport-specific capacity to improve performance and minimise injury risk, but consideration of injury history is also important. In this respect, healthy athletes also benefit from loading to improve local tissue capacity. Conversely, the prescription of training load in injured athletes prioritises the development of local tissue loading to decrease pain and inhibition and increase function. However, local tissue loading in isolation, is insufficient to return injured athletes to competition. Scheduling training to emphasise the development of sport-specific capacity is required to bridge the gap between early stage rehabilitation and return to performance.

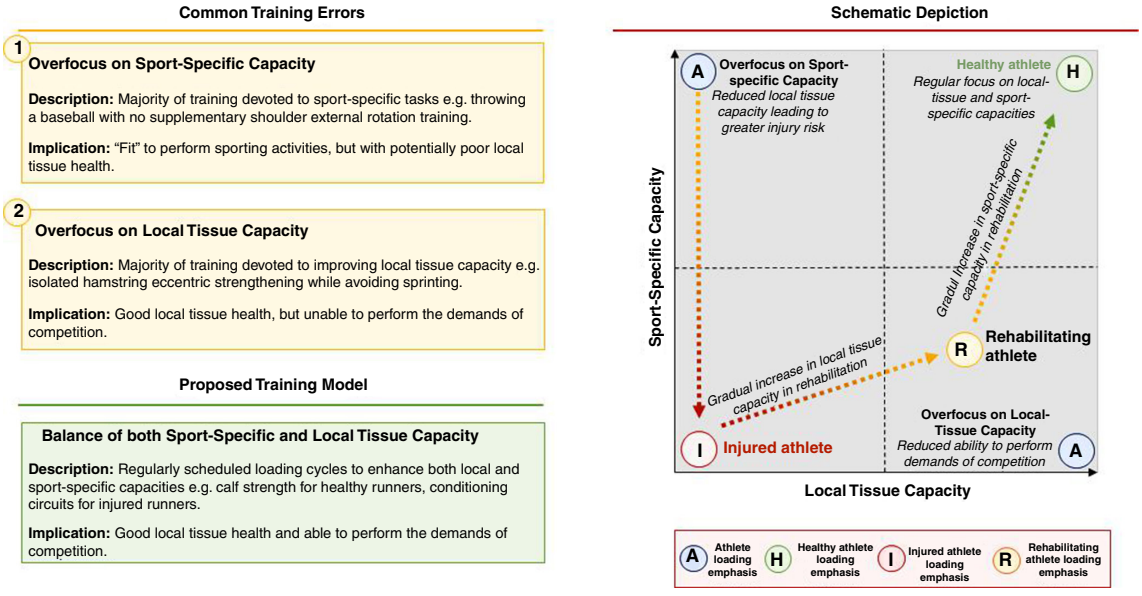

Figure 1 The balance between sport-specific capacity and local-tissue capacity in healthy and injured athletes.

injury is the strongest predictor of subsequent re-injury. Scheduling *local tissue* conditioning within the training cycle maximises the likelihood of healthy athletes remaining injury-free.

3. *Local tissue* and *sport-specific* loading are needed for an injured athlete to return to sport safely.
   Restoring *local tissue* capacity of *both injured and associated structures* is required for injured athletes. However, without regular doses of load designed to improve *sport-specific* capacity, athletes are at risk of being underprepared for the demands of competition on return to sport. Within the constraints of biological healing, exposure to *sport-specific* loading allows safe re-integration into training and competition, while minimising the risk of re-injury.

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